Secretary Problem Research

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Contents

Lecture 1: abstract error SP

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1 Abstract Error SP

1.1 Basic Definitions

$i \approx j$ means that i and j are in-distinguishable in the first observation.

- (reflexivity) $i \approx i$
- (symmetry) $i \approx j \Leftrightarrow j \approx i$
- (connectedness) $i \approx j, r_i < r_j \Rightarrow \forall k, l[r_k, r_l \in [r_i, r_j] \Rightarrow r_k \approx r_l]$

1.2 Algorithmic Representations

Algorithm 1 Oct 14 2021

```
for $i \in [n]$ do
    $\alpha \leftarrow$ the current item to be decided
    if $i < r$ then                                          ▷ Phase 1
        if $\exists \beta \in O_i. \alpha <_i \beta$ then    ▷ $\alpha$ is not a maximum
            Decline this item
        else if $\alpha \in E_i$ then
            Decline this item
        else
            Examine this item
        end if
    else if $i < n$ then                                     ▷ Phase 2
        if $\exists \beta \in O_i. \alpha <_i \beta$ then
            Decline this item
        else
            if $\forall \beta \in O_i. \beta \leq_i \alpha$ then   ▷ $\alpha$ is the absolute maximum
                Accept this item
            else if $\alpha \in E_i$ then
                Decline this item
            else
                Examine this item
            end if
        end if
    else
        Accept this item
    end if
end for
```

Algorithm 2 Game Process, Relaxed

$[n], r_i$ and $\approx$ relation are given according to definition
$O_1 = \{1\}$
$E_1 = \emptyset$
$\alpha_1 = 1$
$\Theta_1 = \emptyset$
for $i \in [n]$ **do**
    $d_i \leftarrow$ player's decision
    if $d_i =$ Accept **then**
        Game terminates with reward $v(\alpha_i)$
    else if $d_i =$ Decline **then**
        $O_{i+1} \leftarrow O_i \cup \{i+1\}$
        $E_{i+1} \leftarrow E_i$
        $\alpha_{i+1} \leftarrow i+1$
    else if $d_i =$ Examine **then**
        $O_{i+1} \leftarrow O_i$
        $E_{i+1} \leftarrow E_i \cup \{i\}$
        $\alpha_{i+1} \leftarrow \alpha$
    end if
    $\Theta_{i+1} \leftarrow \{(\beta,\gamma) \in O_{i+1}^2 : r_\beta \leq r_\gamma, \beta \not\approx \gamma\} \cup \{(\beta,\gamma) \in E_{i+1}^2 : r_\beta \leq r_\gamma\}$
end for

1.3 Calculation under a relaxed condition

We give the expected value of ALG on the relaxed model (only one examine is required to obain precise comparisons). Let α denote the selected item.

$$\mathrm{E}[\mathrm{OPT}] = m \tag{1}$$

$$\mathrm{E}[\mathrm{ALG}] = \sum_{t=1}^{m} t\,\mathrm{P}[v(\alpha) = t] \tag{2}$$

$$\begin{aligned} \mathrm{P}[v(\alpha) = t] = &\sum_{i=r}^{n} \mathrm{P}[\alpha = i \mid v(i) = t]\,\mathrm{P}[v(i) = t] &(3)\\ &+ \mathrm{P}[\text{reach } n \cap \text{not accept } n]\,\mathrm{P}[v(n) = t] &(4)\\ &+ \mathrm{P}[\text{reach } n-1 \cap \text{examine and decline } n-1]\,\mathrm{P}[v(n-1) = t] &(5) \end{aligned}$$

The two added terms are cases where the items are exhausted. If n is reached then it must be selected. If $n-1$ is examined then it must be selected also.

$$\mathrm{P}[v(i) = t] = 1/m, \forall i \forall t \tag{6}$$

$$\begin{aligned}
\mathrm{P}[\alpha = i \mid v(i) = t] &= \mathrm{P}[\text{reach } i \mid \max_{j<r} v(j) \le t]\, \mathrm{P}[\max_{j<r} v(j) \le t] && (7)\\
&= \sum_{s=1}^{t} \mathrm{P}[\text{reach } i \mid \max_{j<r} v(j) = s]\, \mathrm{P}[\max_{j<r} v(j) = s] && (8)
\end{aligned}$$

$$\mathrm{P}[\text{reach } n \cap \text{not accept } n] = \sum_{s=t+1}^{m} \mathrm{P}[\text{reach } n \mid \max_{j<r} v(j) = s]\, \mathrm{P}[\max_{j<r} v(j) = s] \tag{9}$$

$$\mathrm{P}[\text{reach } n{-}1 \cap \text{examine and decline } n{-}1] = \sum_{s=t+1}^{m} \mathbf{1}_{s \le t+\delta}\, \mathrm{P}[\text{reach } n{-}1 \mid \max_{j<r} v(j) = s]\, \mathrm{P}[\max_{j<r} v(j) = s] \tag{10}$$

$$\begin{aligned}
\mathrm{P}[\max_{j<r} v(j) = s] &= \mathrm{P}[\max_{j<r} v(j) \le s] - \mathrm{P}[\max_{j<r} v(j) \le s-1] && (11)\\
&= (\frac{s}{m})^{r-1} - (\frac{s-1}{m})^{r-1} && (12)
\end{aligned}$$

$$\begin{aligned}
\mathrm{P}[\text{reach } i \mid \max_{j<r} v(j) = s] &= \sum_{(a_1,\dots,a_k): a_j \in \{1,2\}, \sum a_j = i-r} \prod_{j=1}^{k} \mathbf{1}_{\delta<t} \left(\mathbf{1}_{a_j=1} \frac{s-\delta-1}{m} + \mathbf{1}_{a_j=2} \frac{\delta}{m} \right) + \mathbf{1}_{\delta \ge t} \frac{t-1}{m} && (13)\\
&= \sum_{x=0}^{\lfloor \frac{i-r}{2} \rfloor} \left(\mathbf{1}_{\delta<t} \left(\frac{\delta}{m}\right)^{x} \left(\frac{s-\delta-1}{m}\right)^{i-r-2x} + \mathbf{1}_{\delta \ge t} \left(\frac{t-1}{m}\right)^{x} 0^{i-r-2x} \right) \frac{(i-r-x)!}{x!(i-r-2x)} && (14)
\end{aligned}$$

Explanation: There are $i - r$ steps to go from r to i. Each decline will go 1 step, and each examine+decline will go 2 steps. We calculate the probability to take each possible path and then sum them up. In the second equation, x is the number of items examined going from r to i.

Lecture 2: Relaxed SP

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1.4 relaxed model PI

1.4.1 calculation 1 (in meeting)

$$\begin{aligned}
&\sum_{i=1}^{n} \mathrm{P}[\text{accept } i \cap i \text{ best}] \\
&= \sum_{i=1}^{n} \mathrm{P}[\text{accept } i \mid i \text{ best}]\, \mathrm{P}[i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^{n} \mathrm{P}[X_1, \ldots, X_{i-1} < T \cap X_i \geq T \mid i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^{n} \mathrm{P}[X_1, \ldots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}]\, \mathrm{P}[X_i \geq T \mid i \text{ is best}]
\end{aligned}$$

Note that $\mathrm{P}[X_i \geq T \mid i \text{ best}] = 1 - F_{X_{\max}}(T) = 1 - T^n$. Also,

$$\begin{aligned}
&\mathrm{P}[X_1, \ldots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \\
&= \int_T^1 \mathrm{P}[X_1, \ldots, X_{i-1} < T \mid i \text{ best} \cap X_i = x] f_X(x | X_i \geq T \cap i \text{ best})\, dx \\
&= \int_T^1 \mathrm{P}[X_1, \ldots, X_{i-1} < T \mid X_1, \ldots, X_{i-1} < x] f_{X_{\max}}(x | X_{\max} \geq T)\, dx \\
&= \int_T^1 \left(\frac{T}{x}\right)^{i-1} \frac{n x^{n-1}}{1 - T^n}\, dx \\
&= \frac{n T^{i-1}}{1 - T^n} \int_T^1 x^{n-i}\, dx \\
&= \frac{n T^{i-1}}{1 - T^n} \frac{1 - T^{n-i+1}}{n - i + 1}
\end{aligned}$$

Combining all terms,

$$\begin{aligned}
&\sum_{i=1}^{n} \mathrm{P}[\text{accept } i \cap i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^{n} \frac{n T^{i-1}}{1 - T^n} \frac{x^{n-i+1}}{n - i + 1} (1 - T^n) \\
&= \sum_{i=1}^{n} T^{i-1} \frac{1 - T^{n-i+1}}{n - i + 1}
\end{aligned}$$

1.4.2 Calculation 2

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0,1)$. Observed values are $\tilde{X}_i \sim \text{Unif}(X_i - \delta, X_i + \delta)$. We adopt a thresholding scheme for ALG: We examine an item if $\tilde{x}_i > T$ and accept if $x_i > T$.

$$\begin{aligned}
\text{P}[X_{max} \text{ selected}] &= \sum_{i=1}^{n} \text{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
&= \sum_{i=1}^{n} \int_0^1 \text{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^{n} \int_T^1 \text{P}[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^{n} \int_T^1 T^{i-1} x^{n-i} dx \\
&= \sum_{i=1}^{n} \frac{1}{n-i+1} T^{i-1}(1 - T^{n-i+1})
\end{aligned}$$

2 Relaxed SP Prophet Inequality

2.1 Lower Bound of P^{max}

In this section, we provide a lower bound of P^{max}, where

$$P^{\text{max}} = \max_{T \in [0,1]} \sum_{i=1}^{n} \frac{1}{n-i+1} T^{i-1}(1 - T^{n-i+1}).$$

The optimal T^* can be solved by the optimization problem. Let's show the lower bound of $P^{\text{max}}(T^*)$. Let $T = \frac{n}{n+1}$, then we have

$$\begin{aligned}
P^{\text{max}}(T^*) &\geq P^{\text{max}}(T) \\
&= \sum_{i=1}^{n} \frac{1}{n-i+1} T^{i-1}(1 - T^{n-i+1}) \\
&= \sum_{i=1}^{n} \frac{1}{n-i+1} T^{i-1} - \sum_{i=1}^{n} \frac{1}{n-i+1} T^{n} \\
&= T^n \sum_{i=1}^{n} \frac{1}{i}(T^{-i} - 1)
\end{aligned}$$

We take the limit $n \to \infty$, we have

$$\lim_{n \to \infty} T^n \sum_{i=1}^{n} \frac{1}{i}(T^{-i} - 1) = \frac{1}{e} \lim_{n \to \infty} \sum_{i=1}^{n} \frac{1}{i}((1 + \frac{1}{n})^i - 1)$$

Observe that

$$(1+\frac{1}{n})^i - 1 = \sum_{k=1}^{i} \binom{i}{k} \frac{1}{n^k}.$$

Then, we take the sum on i, we have

$$\sum_{i=1}^{n} \frac{1}{i}(1+\frac{1}{n})^i - 1 = \sum_{i=1}^{n} \frac{1}{i} \sum_{k=1}^{i} \binom{i}{k} \frac{1}{n^k}$$

By Fubini Theorem, we have

$$\sum_{i=1}^{n} \frac{1}{i} \sum_{k=1}^{i} \binom{i}{k} \frac{1}{n^k} = \sum_{k=1}^{n} \sum_{i=k}^{n} \frac{1}{i} \binom{i}{k} \frac{1}{n^k}$$

We have

$$\frac{1}{i}\binom{i}{k} = \frac{(i-1)(i-2)...(i-k+1)}{k!}$$

The nominator is a polynomial with order $k-1$ with respect to i.

Exercise 2.1. Show that $\sum_{i=1}^{n} i^k = \frac{1}{k+1} n^{k+1} + O(n^k)$

By this property, we have

$$\lim_{n\to\infty} \sum_{i=k}^{n} \frac{1}{i} \binom{i}{k} \frac{1}{n^k} = \frac{1}{k \cdot k!} \frac{n^k}{n^k} = \frac{1}{k \cdot k!}.$$

Finally, we have the lower bound of $P^{\max}$ is

$$\frac{1}{e} \sum_{k=1}^{n} \frac{1}{k \cdot k!} \approx 0.485.$$

2.2 Upper Bound of $P^{\max}$

Consider there are only two secretaries in the arriving sequence. We have $X_1 \sim \text{Unif}(0,1)$, $X_2 \sim \text{Unif}(0,1)$.

Exercise 2.2. Show that the upper bound of $P^{\max}$ among all deterministic algorithms is $\frac{1}{2}$

Hint: Any deterministic algorithm can only choose X_1 or X_2. If $X_1 = x$, what will happen? Then take integral. Another method is that think about a coupling random arrival instance $Y_1 = X_1$, $Y_2 = \text{Unif}(0,1)$. Y obviously follows the arrival process distribution. When the first customer arrives, any algorithm cannot distinguish the arrival instance is X or Y. Then for the first customer, if the algorithm accepts him, ..., if not accept,....

Exercise 2.3. Show that the upper bound of $P^{\max}$ among all randomized algorithms is $\frac{1}{2}$

Hint: Yao's principle. (or set a probability distribution on X_1 and X_2).

Lecture 3: response to excercises

Mon 07 Feb 2022 00:24

2.3 Responses

2.3.1 Response to (2.1)

Proof. Use induction on n. The base case is trivial:

$$1 = \frac{1}{k+1} + O(1).$$

Now suppose the property holds for n.

$$\sum_{i=1}^{n+1} i^k = (n+1)^k + \sum_{i=1}^{n} i^k = (n+1)^k + \frac{1}{k+1} n^{k+1} + O(n^k)$$

$$\frac{1}{k+1}(n+1)^{k+1} = \frac{1}{k+1} n^{k+1} + O(n^k).$$

By considering $(n+1)^k = O(n^k)$ and $O(n^k) \pm O(n^k) = O(n^k)$, the difference of the right hand side of the above equations is $O(n^k)$. The left hand side shows the property for $n+1$. □

2.3.2 Response to (2.2)

Optimal P for a determininstic algorithm is 0.75.

Proof. ALG is competely characterized by a function $\alpha : [0,1] \to [0,1]$, where $\alpha(x)$ represents the probability of accepting X_1 after an observation $X_1 = x$.

$$\begin{aligned}
\mathrm{P}[\mathrm{win}] &= \int_0^1 \alpha\, \mathrm{P}[X_2 < x] + (1-\alpha)\, \mathrm{P}[X_2 > x] f(x)\, dx \\
&= \int_0^1 \alpha x + (1-\alpha)(1-x) \\
&= \int_0^1 2\alpha x - \alpha\, dx + \frac{1}{2} \\
&\leq \max_{\alpha} \int_0^1 2\alpha x - \alpha\, dx + \frac{1}{2}.
\end{aligned}$$

ALG is deterministic iff $\alpha(x) \in \{0,1\}$. If α takes 0, the integral is zero. If α takes 1, $2\alpha x - a > 0$ iff $x > \frac{1}{2}$. Hence to maximize the integral we take $\alpha(x) = \begin{cases} 1 & x \geq \frac{1}{2} \\ 0 & x < \frac{1}{2} \end{cases}$, which gives an upper bound of $\frac{3}{4}$. □

Response to (2.3)

Lecture 4: Relaxed SP with error

Tue 01 Feb 2022 00:06

3 Calculation with error involved

The weights $X_1, \ldots, X_n \sim \mathrm{Unif}(0,1)$. The noisy observations $\tilde{X_1}, \ldots, \tilde{X_n} \sim \mathrm{Unif}(x_i - \delta, x_i + \delta)$.

The strategy is to introduce thresholding parameters R and $T \in (0,1)$. We examine the upcoming item i if $\tilde{x}_i > R$, and accept the item if $x_i > T$. We still want to find $\mathrm{P}[X_{max}$ is selected$]$ given particular R, T, n, δ.

$$\mathrm{P}[X_{max} \text{ selected}] = \sum_i \int_0^1 \mathrm{P}[X_i \text{best} \cap i \text{ selected} \mid X_i = x] f(x)\, dx \tag{15}$$

$$\begin{aligned}
&\mathrm{P}[X_i \text{best} \cap i \text{ selected} \mid X_i = x] \\
&= \mathrm{P}[\text{reach } i \cap \tilde{X}_i > R \cap X_i > T \cap X_i \text{ max} \mid X_i = x] \\
&= \mathrm{P}[\mathrm{Unif}(x-\delta, x+\delta) > R] \mathbf{1}(x > T)\, \mathrm{P}[\text{reach } i \cap X_i \,\text{max} \mid X_i = x]
\end{aligned}$$

$$\begin{aligned}
&\mathrm{P}[\text{reach } i \cap X_i \,\text{max} \mid X_i = x] \\
&= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \mathrm{P}[\text{reach } i \cap X_i \,\text{max} \cap k \text{ examines before reaching } i \mid X_i = x] \\
&= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \mathrm{P}[\text{reach } i \text{ with } k \text{ examines}]\, \mathrm{P}[X_i \text{ max} \mid \text{reach } i \text{ with } k \text{ examines} \cap X_i = x] \\
&= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \underbrace{\frac{(i-1-k)!}{k!\,(i-1-2k)!}}_{\text{ways to reach } i} \mathrm{P}[\mathrm{ED}]^k\, \mathrm{P}[\mathrm{D}]^{i-1-2k} \underbrace{F(x|X<T)^k}_{\text{examined}} \underbrace{F(x)^k}_{\text{omitted}} \underbrace{F(x|\tilde{X}<R)^{i-1-2k}}_{\text{declined directly}} \underbrace{F(x)^{n-i}}_{\text{upcoming}}
\end{aligned}$$

where

$$\begin{aligned}
\mathrm{P}[\mathrm{ED}] &= \frac{1}{2\delta} \int_0^T \int_{y-\delta}^{y+\delta} \mathbf{1}(z > R)\, dz\, dy = \text{const} \\
\mathrm{P}[D] &= \frac{1}{2\delta} \int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R)\, dz\, dy = \text{const}
\end{aligned}$$

and for $0 \le x \le 1$:

$$\begin{aligned}
F(x|X < T) &= \begin{cases} \frac{x}{T} & 0 \le x < T \\ 1 & x \ge T \end{cases} \\
F(x|\tilde{X} < R) &= \frac{\int_0^x \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R)\, dz\, dy}{\int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R)\, dz\, dy} \\
F(x) &= x.
\end{aligned}$$